

Experiment / Case Study No.: 8

1. Experiment Title:

Access and Validate Form Fields Using Events; Design a Gym Admission Form with Live Input Checks

2. Software/Tools Required:

Visual Studio Code, HTML5, CSS3, JavaScript (DOM Access, input/change/blur/submit Events), Google Chrome Browser

3. Program Code — Version 1 (Card-Style Gym Admission Form):

index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Gym Admission Form</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <div class="container">
    <h2>🏋️ Gym Admission Form</h2>

    <form id="gymForm">

      <label for="name">Full Name</label>
      <input type="text" id="name" name="name" placeholder="Enter your full
name" autocomplete="off">
      <span id="nameError" class="error"></span>

      <label for="age">Age</label>
      <input type="number" id="age" name="age" placeholder="Enter your age"
min="1" max="120">
      <span id="ageError" class="error"></span>

      <label for="email">Email</label>
      <input type="email" id="email" name="email"
placeholder="you@example.com" autocomplete="email">
      <span id="emailError" class="error"></span>

      <label for="mobile">Mobile Number</label>
      <input type="tel" id="mobile" name="mobile" placeholder="10-digit
mobile number" maxlength="10">
```

```

        <span id="mobileError" class="error"></span>

        <label for="plan">Membership Plan</label>
        <select id="plan" name="plan">
            <option value="">-- Select Plan --</option>
            <option value="monthly">Monthly</option>
            <option value="quarterly">Quarterly</option>
            <option value="yearly">Yearly</option>
        </select>
        <span id="planError" class="error"></span>

        <button type="submit" id="submitBtn">
            <span>Join Now</span>
            <div class="spinner" id="spinner"></div>
        </button>

        <p id="result" class="success"></p>

    </form>
</div>

<script src="script.js"></script>
</body>

</html>

```

script.js

```

document.addEventListener('DOMContentLoaded', function () {
    const form = document.getElementById('gymForm');
    const nameInput = document.getElementById('name');
    const ageInput = document.getElementById('age');
    const emailInput = document.getElementById('email');
    const mobileInput = document.getElementById('mobile');
    const planSelect = document.getElementById('plan');

    const nameError = document.getElementById('nameError');
    const ageError = document.getElementById('ageError');
    const emailError = document.getElementById('emailError');
    const mobileError = document.getElementById('mobileError');
    const planError = document.getElementById('planError');
    const result = document.getElementById('result');
    const submitBtn = document.getElementById('submitBtn');
    const spinner = document.getElementById('spinner');

    nameInput.addEventListener('input', function () {
        const name = this.value.trim();

        if (name === '') {
            nameError.textContent = 'Name is required';
            this.style.borderColor = '#e53e3e';
        } else if (!/^[A-Za-z\s]+$/.test(name)) {

```

```

        nameError.textContent = 'Only letters and spaces allowed';
        this.style.borderColor = '#e53e3e';
    } else {
        nameError.textContent = '';
        this.style.borderColor = '#e2e8f0';
    }
});

ageInput.addEventListener('blur', function () {
    const age = parseInt(this.value);

    if (isNaN(age) || age === '') {
        ageError.textContent = 'Please enter a valid age';
        this.style.borderColor = '#e53e3e';
    } else if (age < 1 || age > 150) {
        ageError.textContent = 'Age must be between 1 and 150';
        this.style.borderColor = '#e53e3e';
    } else if (age < 16 || age > 60) {
        ageError.textContent = 'Age must be between 16 and 60';
        this.style.borderColor = '#e53e3e';
    } else {
        ageError.textContent = '';
        this.style.borderColor = '#e2e8f0';
    }
});

ageInput.addEventListener('input', function () {
    const age = parseInt(this.value);
    if (!isNaN(age) && age >= 16 && age <= 60) {
        ageError.textContent = '';
        this.style.borderColor = '#48bb78';
    } else if (this.value !== '') {
        this.style.borderColor = '#e53e3e';
    }
});

emailInput.addEventListener('input', function () {
    const email = this.value.trim();
    const pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

    if (email === '') {
        emailError.textContent = 'Email is required';
        this.style.borderColor = '#e53e3e';
    } else if (!pattern.test(email)) {
        emailError.textContent = 'Please enter a valid email';
        this.style.borderColor = '#e53e3e';
    } else {
        emailError.textContent = '';
        this.style.borderColor = '#e2e8f0';
    }
});

```

```

mobileInput.addEventListener('input', function () {
    const mobile = this.value.trim();

    // Remove any non-digit characters for validation
    const digitsOnly = mobile.replace(/\D/g, '');

    if (mobile === '') {
        mobileError.textContent = 'Mobile number is required';
        this.style.borderColor = '#e53e3e';
    } else if (digitsOnly.length < 10) {
        mobileError.textContent = 'Enter at least 10 digits';
        this.style.borderColor = '#e53e3e';
    } else if (digitsOnly.length > 10) {
        mobileError.textContent = 'Mobile number must be exactly 10 digits';
        this.style.borderColor = '#e53e3e';

        this.value = digitsOnly.slice(0, 10);
    } else {
        mobileError.textContent = '';
        this.style.borderColor = '#48bb78';
    }
});

planSelect.addEventListener('change', function () {
    if (this.value === '') {
        planError.textContent = 'Please select a membership plan';
    } else {
        planError.textContent = '';
    }
});

form.addEventListener('submit', function (e) {
    e.preventDefault();

    const hasErrors = nameError.textContent !== '' ||
        ageError.textContent !== '' ||
        emailError.textContent !== '' ||
        mobileError.textContent !== '' ||
        planError.textContent !== '';

    if (hasErrors) {
        const firstError = form.querySelector('.error:not(:empty)');
        if (firstError) {
            firstError.scrollIntoView({ behavior: 'smooth', block: 'center' });
        }
        return;
    }
}

```

```

submitBtn.classList.add('loading');
submitBtn.disabled = true;

setTimeout(function () {
    result.style.display = 'block';
    result.textContent = '✅ Gym Admission Successful! Welcome aboard!';

    form.reset();
    nameError.textContent = '';
    ageError.textContent = '';
    emailError.textContent = '';
    mobileError.textContent = '';
    planError.textContent = '';

    submitBtn.classList.remove('loading');
    submitBtn.disabled = false;

    setTimeout(function () {
        result.style.display = 'none';
    }, 5000);
}, 1500);
});
});

```

4. Output — Version 1:

Fig 4: Card-Style Gym Admission Form output — PRN: 24070521062, Name: Soumya Suryawanshi, File Path: C:/Users/soumya suryawanshi/OneDrive/Pictures/java practicals/java practical 208/index.html

5. Program Code — Version 2 (Table-Style Gym Admission Form):

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Gym Admission Registration Form</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <form id="gymForm">
    <table>
      <thead>
        <tr>
          <th class="col-name">Name</th>
          <th class="col-value">Value</th>
        </tr>
      </thead>
      <tbody>
        <tr>
          <td>Name</td>
          <td><input type="text" id="name" name="name"></td>
        </tr>
        <tr>
          <td>Sex</td>
          <td>
            <label><input type="radio" name="sex" value="Male">
Male</label><br>
            <label><input type="radio" name="sex" value="Female"
checked> Female</label>
          </td>
        </tr>
        <tr>
          <td>Eye color</td>
          <td>
            <select id="eyeColor" name="eyeColor">
              <option value="green">green</option>
              <option value="brown">brown</option>
              <option value="blue">blue</option>
              <option value="hazel">hazel</option>
            </select>
          </td>
        </tr>
        <tr>
          <td>Check all that apply</td>
          <td>
            <label><input type="checkbox" name="height" value="Over 6
feet tall"> Over 6 feet tall</label><br>
            <label><input type="checkbox" name="weight" value="Over 200
pounds"> Over 200 pounds</label>
          </td>
        </tr>
      </tbody>
    </table>
  </form>
</body>
</html>
```

```

        <tr>
            <td colspan="2" class="desc-label">Describe your athletic
ability:</td>
        </tr>
        <tr>
            <td colspan="2" class="textarea-cell">
                <textarea id="ability" name="ability"></textarea>
            </td>
        </tr>
        <tr>
            <td colspan="2" class="submit-cell">
                <input type="submit" value="Enter my information">
            </td>
        </tr>
    </tbody>
</table>
</form>
<script src="script.js"></script>
</body>
</html>

```

script.js

```

document.addEventListener("DOMContentLoaded", function() {
    const form = document.getElementById("gymForm");
    const nameInput = document.getElementById("name");

    // Example of onblur event for real-time validation feedback
    nameInput.addEventListener("blur", function() {
        if (this.value.trim() === "") {
            this.style.backgroundColor = "#ffcccc"; // Light red if empty
        } else {
            this.style.backgroundColor = "white"; // Reset to white
        }
    });

    // Handle form submission event for validation
    form.addEventListener("submit", function(event) {
        let isValid = true;
        let errorMessage = "";

        const name = nameInput.value.trim();
        const ability = document.getElementById("ability").value.trim();

        // Validation check for Name
        if (name === "") {
            errorMessage += "Name field cannot be empty.\n";
            isValid = false;
        }

        // Validation check for Athletic Ability
        if (ability === "") {
            errorMessage += "Please describe your athletic ability.\n";
        }
    });
});

```

```

        isValid = false;
    }

    // If not valid, show error and prevent submission
    if (!isValid) {
        alert(errorMessage);
        event.preventDefault(); // Stop form submission
    } else {
        // Form is valid
        alert("Registration successful!\nWelcome to the Gym, " + name + "!");
        event.preventDefault(); // Stop actual page reload for demonstration
        form.reset(); // Reset form fields
        nameInput.style.backgroundColor = "white"; // Reset custom style
    }
});
});

```

6. Output — Version 2:

NAME	VALUE
Name	soumya suryawanshi
Sex	☒ Male ☐ Female
Eye color	green
Check all that apply	☐ Over 6 feet tall ☒ Over 200 pounds
Describe your athletic ability:	
Enter my information	

Fig 6: Table-Style Gym Admission Form output — PRN: 24070521062, Name: Soumya Suryawanshi, File Path: C:/Users/soumya suryawanshi/OneDrive/Pictures/java practicals/java practical 208/new updated gym from/index.html

7. Result/Conclusion:

A Gym Admission Form was successfully designed and implemented in two versions using HTML and JavaScript. In the first (card-style) version, form fields were accessed through the DOM and validated live using input, blur, and change events — checking the name pattern, age range, email format, a 10-digit mobile number, and a selected membership plan — with the submit event blocking submission until all fields passed validation. In the second (table-style) version, the name field was validated on the blur event and the form's submit event checked that the name and athletic-ability fields were not left empty before accepting the registration. Both implementations produced the expected output for the given inputs and reinforced practical understanding of accessing form fields via the DOM and using JavaScript events for live, submission-time validation.

Upload the Soft copy as well as complete code on GitHub link is mandatory.